

Coding Theory and Mathematics

– Linear Code and Cyclic Code –

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수학의 즐거움, *Enjoying Math*

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Contents



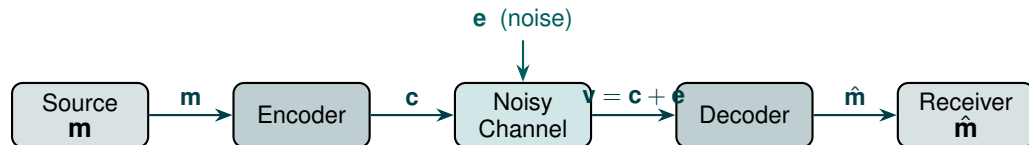
I. Introduction

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The Communication Problem



Setup

- ▶ Message $\mathbf{m} \in \mathbb{F}_q^k$
- ▶ Encoder adds redundancy: $\mathbf{m} \mapsto \mathbf{c} \in \mathbb{F}_q^n$, $n > k$
- ▶ Channel corrupts: $\mathbf{v} = \mathbf{c} + \mathbf{e}$

Goal

- ▶ Recover $\hat{\mathbf{m}} = \mathbf{m}$ even when $\mathbf{e} \neq \mathbf{0}$
- ▶ Keep rate $R = k/n$ as large as possible



I. Introduction

Shannon's Channel Coding Theorem (1948)

Shannon's Channel Coding Theorem

For any discrete memoryless channel with capacity C and any rate $R < C$, there exists a sequence of codes of rate $\rightarrow R$ whose block error probability $\rightarrow 0$.

What it guarantees

- ▶ Reliable communication *is* possible below capacity
- ▶ Error probability can be made arbitrarily small

What it does not provide

- ▶ *How* to construct such codes explicitly
- ▶ *How* to decode efficiently

The Central Challenge of Coding Theory

Construct **explicit algebraic codes** that are both **efficient** (high rate) and **powerful** (correct many errors).



I. Introduction

Roadmap

II. Linear Codes

Structure via **linear algebra**

- Subspaces of $\mathbb{F}_q^n$
- Generator G , parity-check H
- Hamming distance & error correction

III. Cyclic Codes

Structure via **polynomial rings**

- Quotient $\mathbb{F}_q[X]/(X^n - 1)$
- Generator polynomial $g(X)$, ideals
- Hardware-efficient encoding

IV. Applications

CRC	Error detection (Ethernet, USB, ZIP)
BCH / RS	Error correction (CD, QR, deep-space)
Post-quantum	HQC cryptosystem (NIST PQC standard)

Key Observation

Cyclic codes combine the elegance of **abstract algebra** with the efficiency of **shift-register circuits**.